

0. The goal of phase 2b is to perform benchmarking/scalability tests of sample three-tier lakehouse solution.

1. In main.tf, change machine_type at:

```
module "dataproc" {
  depends_on    = [module.vpc]
  source        = "github.com/bdg-tbd/tbd-workshop-1.git?ref=v1.0.36/modules/dataproc"
  project_name  = var.project_name
  region        = var.region
  subnet        = module.vpc.subnets[local.notebook_subnet_id].id
  machine_type  = "e2-standard-2"
}
```

and substitute "e2-standard-2" with "e2-standard-4".

Zmienione wartości w pliku main.tf:

```
module "dataproc" {
  depends_on    = [module.vpc]
  source        = "./modules/dataproc"
  project_name  = var.project_name
  region        = var.region
  subnet        = module.vpc.subnets[local.notebook_subnet_id].id
  machine_type  = "e2-standard-4"
}
```

2. If needed request to increase cpu quotas (e.g. to 30 CPUs):

<https://console.cloud.google.com/apis/api/compute.googleapis.com/quotas?project=tbd-2023z-9918>

Zwiekszona wartość limitu procesorów:

Hello,

Your quota request for tbd-20001-314241 has been approved and your project quota has been adjusted according to the following requested limits:

NAME	DIMENSIONS	REGION	REQUESTED LIMIT	APPROVED LIMIT
CPU	region=europe-west1	europe-west1	32	32
CPU_ALL_REGIONS		GLOBAL	32	32

After approval, Quotas can take up to 15 min to be fully visible in the Cloud Console and available to you.

To verify, please navigate to <https://console.cloud.google.com/iam-admin/quotas?project=tbd-20001-314241>.

If you find actual limits are greater than expected, this is normal if previous requests were approved with higher limits. Otherwise, please let us know if approved changes are not reflected in your project.

If you want to increase your quota further, please file a new request.

Best regards and happy computing!

Sincerely,
Google Cloud Support

[←](#) Szczegóły interfejsu API / usługi ■ WYŁĄCZ INTERFEJS API

WSKAŹNIKI

LIMITY PRZYDZIAŁU I LIMITY SYSTEMU

DANE UWIERZYTELNIAJĄCE

KOSZT

Aktualizuję wartości limitów. Może to potrwać od 2 do 3 tygodni.

DOWIEDZ SIĘ WIĘCEJ

Bieżące wykorzystanie > 90%

1

[Wyświetl limity systemu](#)

Szczytowe wykorzystanie w ciągu 7 dni > 90%

1

[Wyświetl limity systemu](#)

Wszystkie limity systemu

14 022

ZARZĄDZAJ ZASADAMI TWORZENIA ALERTÓW

Filtr

Wpisz nazwę lub wartość właściwości

☐	Nazwa	Typ	Wymiary (np. lokalizacja)	Wartość	Procent bieżącego wykorzystania ↓	Obecne wykorzystanie	Możesz zmienić	
☐	Persistent Disk SSD (GB)	Limit	region : europe-west1	500 GB	100%	500 GB	Tak	
☐	CPUs	Limit	region : europe-west1	32	65,63%	21	Tak	

3. Using tbd-tpc-di notebook perform dbt run with different number of executors, i.e., 1, 2, and 5, by changing:

```
"spark.executor.instances": "2"
```

in profiles.yml.

4. In the notebook, collect console output from dbt run, then parse it and retrieve total execution time and execution times of processing each model. Save the results from each number of executors.

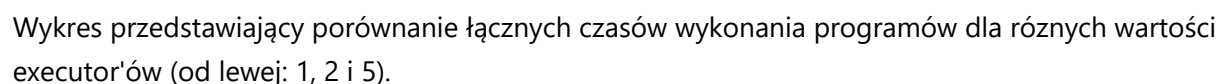
Na poniższej tabeli porównano wyniki czasowe wykonania dla każdego z modeli, porównując ze sobą ilości executor'ów (1, 2, 5).

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		Executors = 1	Executors = 2	Executors = 5
Lp.	Tabela	Czas [s]	Czas [s]	Czas [s]
1	demo_bronze.brokerage_cash_transaction	44,35	37,32	39,45
2	demo_bronze.brokerage_daily_market	145,24	111,24	100,45
3	demo_bronze.brokerage_holding_history	17,41	19,69	17,99
4	demo_bronze.brokerage_trade	67,52	44,78	38,51
5	demo_bronze.brokerage_trade_history	65,33	55,77	48,21
6	demo_bronze.brokerage_watch_history	71,02	65,83	53,71
7	demo_bronze.crm_customer_mgmt	12,33	14,52	17,13
8	demo_bronze.finwire_company	7,66	5,36	4,21
9	demo_bronze.finwire_financial	78,49	69,41	52,74
10	demo_bronze.finwire_security	5,45	6,02	4,96
11	demo_bronze.hr_employee	5,77	5,98	4,25
12	demo_bronze.reference_date	3,45	3,31	4,98
13	demo_bronze.reference_industr	2,98	2,74	2,64
14	demo_bronze.reference_status_type	3,04	3,88	3,51
15	demo_bronze.reference_tax_rate	3,01	4,08	3,96
16	demo_bronze.reference_trade_type	2,88	3,02	2,82
17	demo_bronze.syndicated_prospect	7,12	7,96	6,19
18	demo_silver.daily_market	3345,11	752,73	483,17
19	demo_silver.employees	7,63	13,55	2,92
20	demo_silver.date	5,96	6,11	5,42
21	demo_silver.companies	10,44	9,81	10,23
22	demo_silver.accounts	17,41	16,33	14,99
23	demo_silver.customers	14,59	14,03	13,02
24	demo_silver.trades_history	559,11	278,04	129,42
25	demo_gold.dim_broker	7,47	7,12	6,64
26	demo_gold.dim_date	5,39	4,93	4,08
27	demo_gold.dim_company	5,58	5,02	4,17
28	demo_silver.financials	123,79	88,39	68,65
29	demo_silver.securities	8,03	9,44	7,87
30	demo_silver.cash_transactions	91,08	59,46	50,25
31	demo_gold.dim_customer	21,77	20,36	16,54
32	demo_gold.dim_trade	346,71	241,97	164,89
33	demo_silver.trades	412,89	306,91	325,06
34	demo_gold.dim_security	5,86	6,02	10,44
35	demo_silver.watches_history	200,77	112,61	310,89
36	demo_gold.dim_account	14,55	16,23	20,89
37	demo_silver.holdings_history	197,67	91,07	120,42
38	demo_silver.watches	414,92	255,45	272,37
39	demo_gold.fact_cash_transactions	288,79	224,76	170,39
40	demo_gold.fact_trade	1103,88	743,66	246,92
41	demo_gold.fact_holdings	1896,37	1243,72	893,14
42	demo_gold.fact_watches	681,37	478,19	350,47
43	demo_gold.fact_cash_balances	275,13	211,56	201,56
SUM		10605,32	5678,38	4310,52

5. Analyze the performance and scalability of execution times of each model. Visualize and discuss the final results.

Wykres czasu wykonania dla wszystkich tabel dla różnych executor'ow.



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po między czasami wykonania dla $\text{executor}'a = x$ i $\text{executor}'a = x + 1$ jest coraz mniejsza, co świadczy o tym, że im większa liczba executor'ów, tym coraz to mniejszy poprawy wydajności czasowej.